

Assembly is nearly free. So why does almost no one capture the value?



Weeks of work → hours ^[1]

Anthropic internal experience

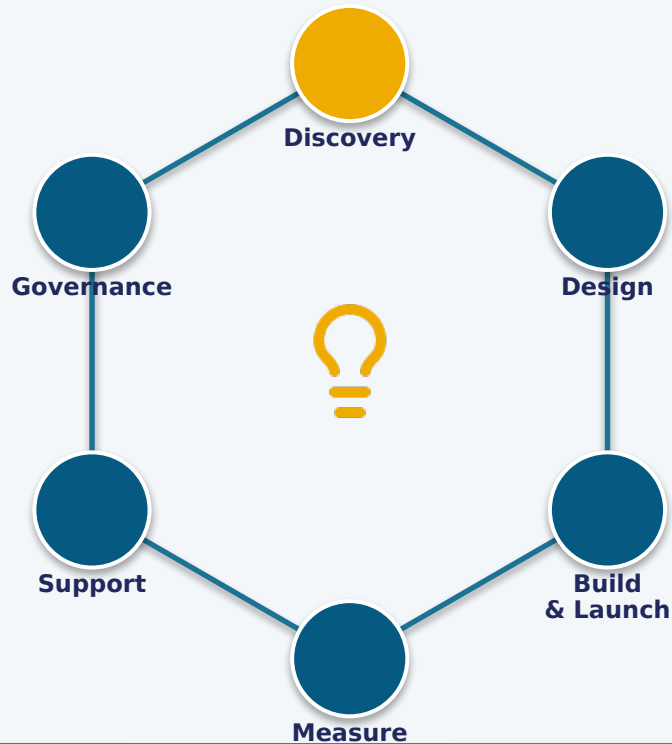


~95% of pilots — zero return ^[2]

MIT, summer 2025

Both facts are true at the same time: assembly became nearly free, but turning assembly into value did not. Hold the question — we'll come back to it at the end of the lecture.





LECTURE 5

AI Product: the full lifecycle — from intent to operation

Course "Deliberate Use of AI" · 3rd-year engineering students

Six sections — six arrows of one loop: code (Lecture 4) is one cheap step inside it.

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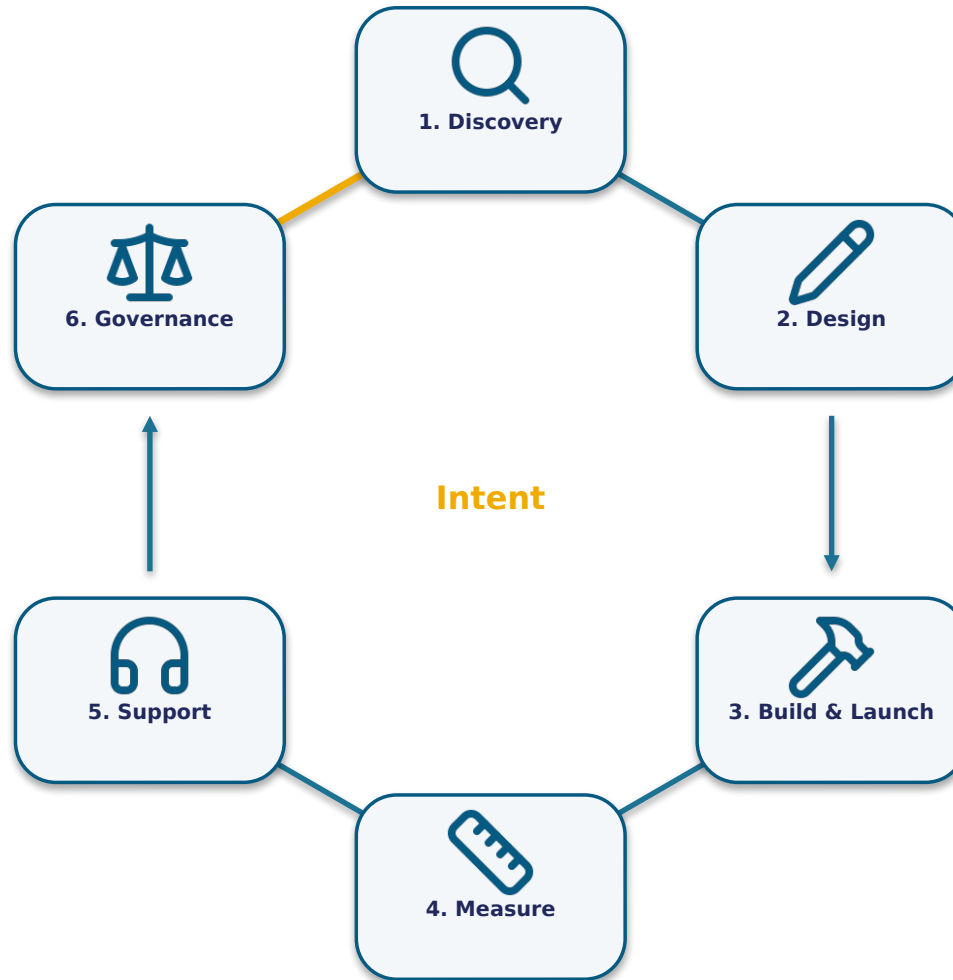
3
Build & Launch

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Measure

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Support

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Governance

The lecture is a loop, not a list of six topics



1. Discovery — where the hypothesis comes from

2. Design — hypothesis → artifact

3. Build & Launch — artifact → product

4. Measure — did it work

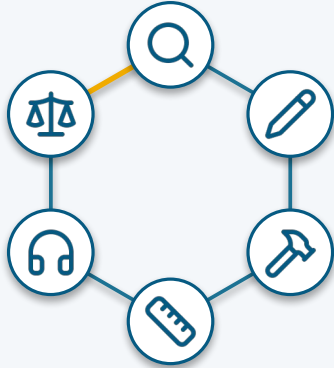
5. Support — lives 24/7

6. Governance — where to invest

The return arrow: governance feeds discovery again — a closed loop, not a one-time linear process.

Lecture 4's code loop is one step inside the product loop

Product loop (Lecture 5)



6 phases



discovery → design → build/launch → measure → operate → govern → discovery again

Code loop (Lecture 4)

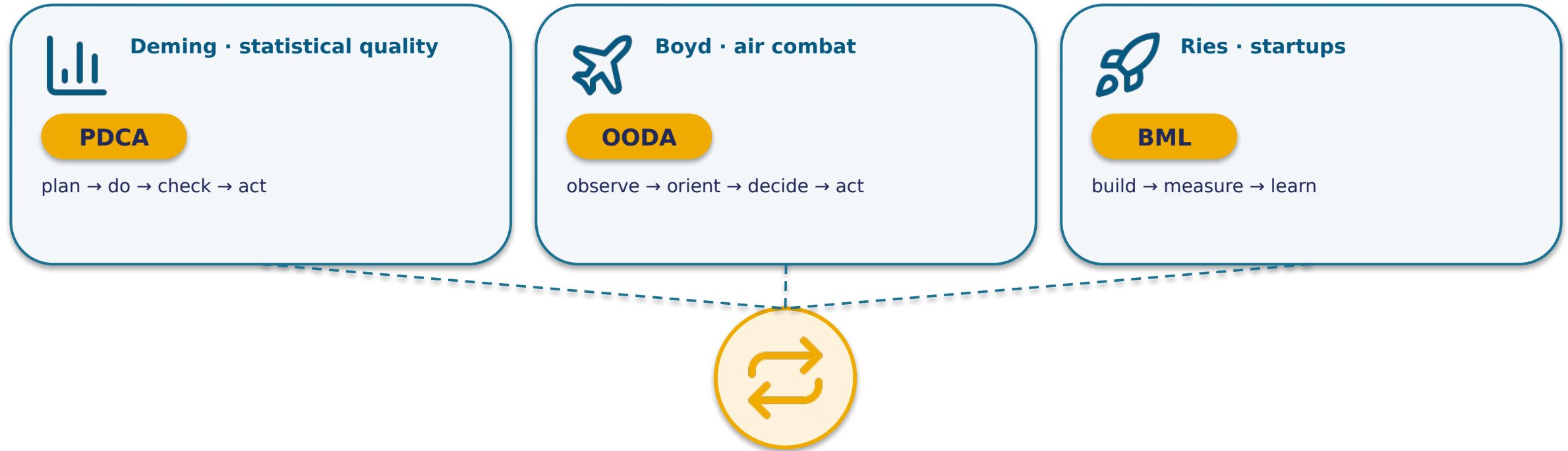
spec → *ADR* → *plan* → *PR* → *incident*



Now that AI made assembly nearly free — what became the product's real bottleneck, and on every phase of the loop: which classical discipline stays, what does AI speed up, and where does AI-first break?

Lecture 4's unit of work is the pull request. The unit of work here is a hypothesis, an experiment, a release decision.

Three people from different fields drew the same blueprint



Without coordinating — the same blueprint: the feedback loop as a structural property of any system that learns under uncertainty.

AI changes the cost and trust of every arrow of the loop — but not equally



Build

cost nearly collapsed to zero



Measure / Learn

cost about the same, trust fell



Observe / Orient

faster, but more exposed to attack

A meta-pattern repeating in each of the 6 sections: every arrow has a classical discipline — AI doesn't cancel it, it changes its cost and the degree of verification it requires.

Think about it: on which arrow of your latest task did speed outrun your actual trust in the result?

SECTION 1

Discovery — intent and hypothesis

2 base cases · 3 failures

Where does a hypothesis about what to build even come from. Most people have the informal experience of "ask your friends" — and almost no one has met the formal discipline that explains why that experience systematically lies.



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Discovery in plain terms



in plain terms

What it is

The phase where you look not for a solution but for a problem: who has the pain, how bad it is, and whether the person is willing to pay for a fix in time, reputation, or money.

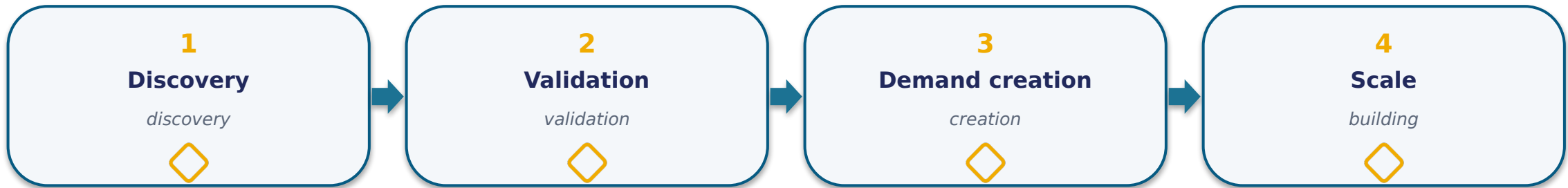
Why it matters

You can build anything; only what solves a real pain for a real person has value. "I'd use something like that" is not a fact.

Mental model

There are no facts inside the building — facts are outside. Write down a hypothesis, go out to people, check it. AI speeds up the gathering, but doesn't replace the conversation with a real person.

"There are no facts inside the building" — 4 steps, each ending in a decision



Steve Blank — the Customer Development methodology ^[1]



Falsifiable hypothesis

we believe X → we'll test it via Y → by date Z we'll have a yes/no answer

BML is the engine; Customer Development is the map of the terrain the engine drives on. First understand what you're testing and with whom — only then how fast to spin the loop.

^[1] [Blank — Customer Development \(Four Steps to the Epiphany\)](#)

Even your mother will lie to you — if the question is built wrong ^[1]



About their life, not your idea

don't pitch the idea and don't ask for a verdict



About the past, not the future

hypothetical questions invite dishonest optimism



A commitment, not a compliment

legitimate signals: time, reputation, money

Bad

"Would you pay \$20 a month for this?"

Good

"How much do you pay today for the closest thing to this?"

Qualitative answers "why" on a small sample and generates hypotheses; quantitative answers "how much" at scale and tests a hypothesis that already exists — not a hierarchy, a division of labor.

Hours → minutes: but links exist so they can be checked



Desk research

Perplexity Deep Research: 2 hours → ~30 minutes ^[1]

verify the source



Interview synthesis

Dovetail — pain-point clustering at scale ^[2]



Reference dataset

a curated collection of "query → correct answer"

97% of researchers use AI — only ~8% trust AI personas as data ^[3]

Mandatory practice: verify important data directly against the source before using it in a decision. Synthetic users are only a pre-research step (piloting a guide, a draft of personas, generating hypotheses), never evidence for a go/no-go decision.

A synthetic interviewee structurally cannot produce disagreement



Mirror of agreement: a synthetic interviewee structurally cannot reflect disagreement — it has no real past that could contradict the way the question is phrased.

When a live interview is strictly better than AI synthesis



1. The cost of a "continue/stop" mistake is high



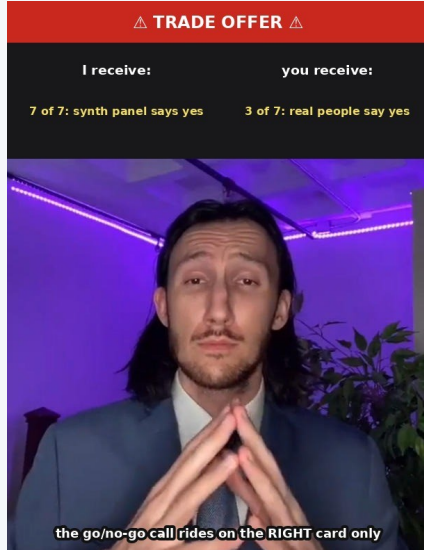
2. You need a real past, not a plausible one



3. You need validation by commitment, not by opinion

AI summaries lose 20-40% of interview detail (Torres) ^[1] when the "individually first" step is skipped — a failure traceable to a specific step, not a vague "AI is sometimes wrong."

The same task: real people — 3 of 7, the synthetic panel — 7 of 7 ^[1]



Real people



3 of 7 — "far-fetched and useless"

Synth panel



7 of 7 — "a game changer"

The same feature — opposite verdicts

Criterion: synthetic output is disqualified for a "continue/stop" decision by construction, not from bad luck on a run. Alternative: a pre-research role + real testing with 5-8 participants.

A \$440,000 government report — with fabricated quotes inside



Deloitte Australia, 2025

- References to nonexistent papers + a fabricated court quote ^[1]
- Produced via Azure OpenAI with no human verification of citations



A\$440,000

~US\$290,000 report

Base rate: ~712 court cases worldwide involving AI-hallucinated citations, ~90% of them in 2025 ^[2]

Lesson: model output during the discovery phase is an unverified draft, not a source of facts. Criterion: a final document for an external client requires verifying 100% of citations, not a sample.

\$62 million — and not a single patient treated



\$62M

0 patients

MD Anderson: partnership closed (2016) ^[1]

IBM Watson for Oncology, since 2012



Internal documents: recommendations were "unsafe and incorrect" ^[1]



Trained on hypothetical cases from a handful of MSK oncologists — not on real outcomes ^[2]

Criterion: a high-stakes domain + only synthetic/hypothetical data behind the recommendation = the product isn't ready, no matter how impressive the demo. Alternative: a system built on evidence-based clinical guidelines with transparent source traceability.

SECTION 2

Design — a hypothesis becomes an artifact

2 base cases · 2 failures

How a hypothesis from research turns into a concrete artifact you can show a person. This is also where safety gets built in — in the design brief, not patched on after something has already gone wrong.



jump straight to the solution

the right problem first

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Design in plain terms



in plain terms

What it is

A hypothesis becomes a cheap draft: a screen sketch, a clickable prototype — something you can show a person.

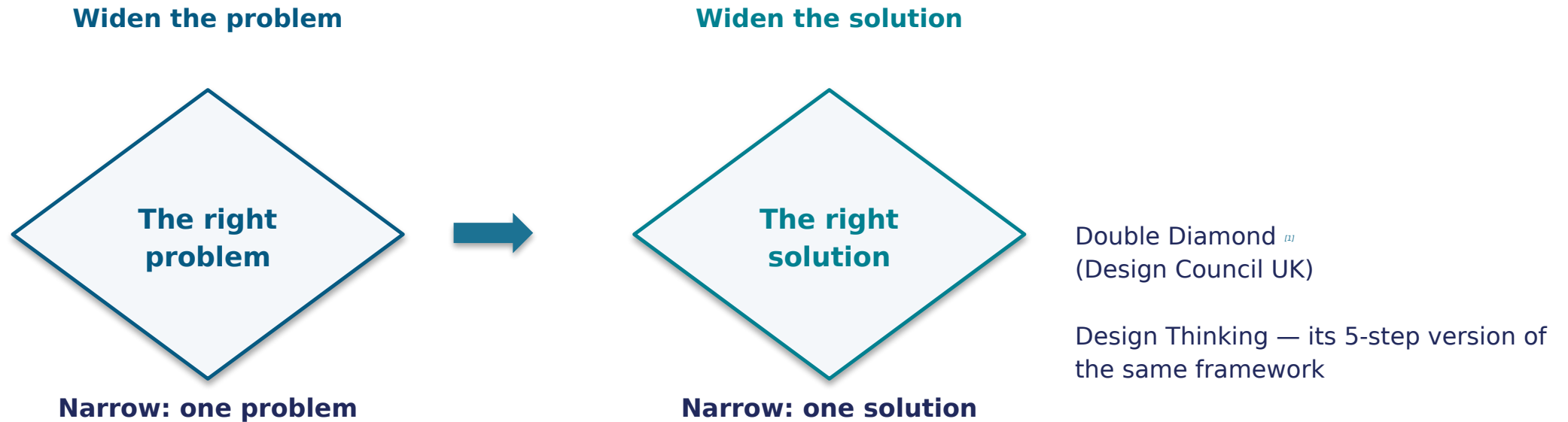
Why

You won't mind throwing away a draft. Discovering "it doesn't work" on paper is cheap; discovering the same thing after launch is very expensive.

Mental model

Two diamonds: first find the right problem, then the right solution. A common mistake is jumping straight to the solution.

Two diamonds: first the right problem, then the right solution



A common mistake: converging on a solution without checking the problem itself — skipping the first diamond. A cheap prototype exists precisely to surface this before money is spent on development.

Nielsen's heuristics are a linter for UX, not a substitute for testing



Top 5 of 10 heuristics ^[1]

- 1 Visibility of system status
- 2 Match with the user's language
- 3 User control and freedom (undo)
- 4 Consistency
- 5 Error prevention



Design system = guardrail

Keeps generative freedom inside an already-validated brand.

Metaphor: heuristics are like a linter for the interface. They catch typical problems before money is spent on research — but don't replace testing with a real user ^[2].

I'm once again asking you: heuristics are a linter, not a live-user test

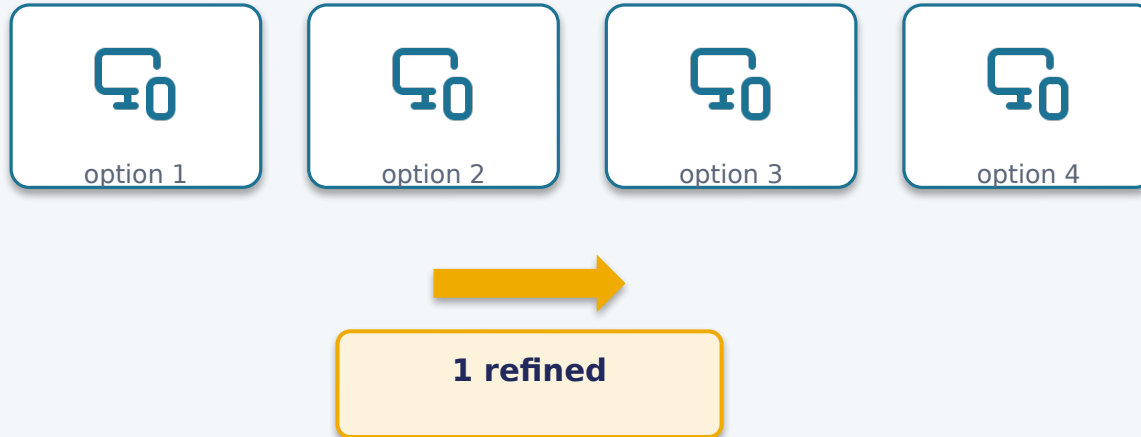


I am once again asking

"I'm once again asking": heuristics catch typical problems, but don't replace testing with a real user.

AI for divergence, humans for convergence

Direction generation



2025-26 tools

v0 (Vercel) · Figma Make ^[1] · Google Stitch ^[2] · bolt.new

- 2-4 directions in minutes (used to be a day of manual wireframing)
- Figma Make pulls in the team's own components — less rework

Convergence needs judgment about specific context that isn't in the training data: AI widens the space of options, choosing and testing with a real user stays with the human.

29% WCAG conformance across 21,880 evaluations — the platform matters more than the prompt ^[1]



21,880 WCAG evaluations of AI-generated interfaces:

- contrast 26.8%
- use of color 19.2%

"Make it accessible" in the prompt does not guarantee the result — platform defaults decide, not the request text.

Design for NON-DETERMINISTIC output: same input → different output → breaks the consistency heuristic; you need human checkpoints.

Safeguards appeared almost two years after launch — only after the tragedy



Protection for vulnerable users was added as a retrofit — after the tragedy, not in the original design brief.



Character.AI

Character.AI

A 14-year-old user died after months of conversation with an AI character (02.2024) ^[1].



Root cause — not a runtime bug, but a missing requirement in the design brief: the MVP optimized for engagement without asking "who could be harmed." Criterion: safeguards for vulnerable users belong in the MVP, not as a patch after a tragedy.

Found by accident: the same application with a different birth date — an instant invite



A filter with a hardcoded "birth date" rule

Women 55+ / men 60+ → auto-rejection

Found by accident: resubmitting the same application with a different birth date → an instant invite.



\$365,000 ^[1]

EEOC, Aug 9, 2023 — the first-ever AI discrimination settlement

Contrast with Character.AI: there, design did NOT build in a safeguard; here, design DID build in a specific harmful rule. Criterion: automating a decision about people → audit for discriminatory features BEFORE release.

^[1] [EEOC \(Aug 9, 2023\) — iTutorGroup](#)

SECTION 3

Build & Launch — the collapsed arrow

2 base cases · 2 failures

This is exactly where AI changes the most — the cost of writing the code itself. But the section doesn't start from that fact — it starts from the classical discipline of release: how to ship safely and when to stop.



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Build and launch in plain terms



in plain terms

What it is

The artifact becomes a working product for users. AI made writing the code itself nearly free.

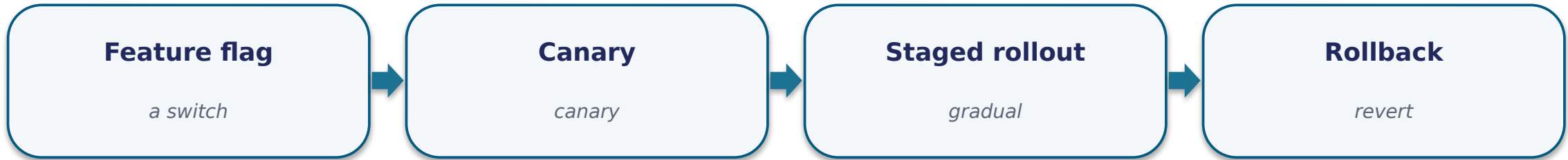
Why release mechanics matter

Since building is cheap, the cost of a mistake is no longer "writing it" but "shipping the wrong thing to everyone at once." You need to turn things on gradually and roll back fast.

Mental model

Launch is a valve, not a switch: 1% → 10% → 100%, watch the reaction. And keep a "kill" decision ready, not just "keep polishing forever."

Same mechanics — but here the decision is to kill the product, not flip a flag



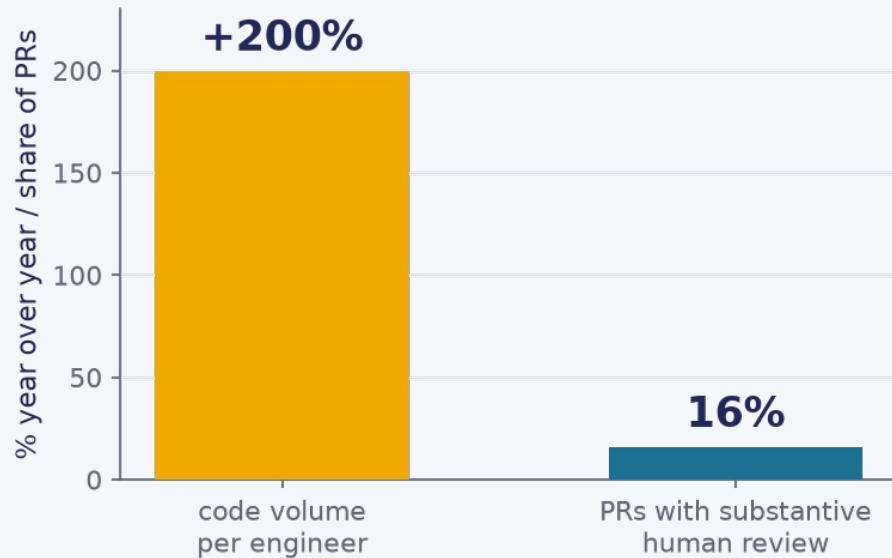
You know these primitives from engineering rollout (CI/CD)



NEW here — whose decision they serve and by what criteria: MVP (Ries) ^[1] = learning, not shipping. Stage-Gate go/kill (Cooper) ^[2] — "a funnel, not a tunnel": failure thresholds are written down as numbers in advance.

The common thread: launch speed is bought with a limited blast radius — how many users are affected if the new thing turns out bad.

+200% code per engineer — but only 16% of PRs got substantive review ^[1]



Anthropic's own measurement

- Implementation: weeks/months → minutes of agentic execution
- Code volume per engineer: +200% year over year
- Substantive human review before merge: only ~16% of PRs

The scarce resource shifted to the two human ends: precision of intent going in and quality of judgment coming out — the bottleneck is now review, not writing.

Launch isn't "flip it on" — it's a staged transfer of control

v3

Automatic, with fallback to a human

earned by track record

v2

Proposes solutions for human approval

human in the loop

v1

High control / low agency

routes

CC/CD vs the familiar CI/CD ^[1]

CC/CD — Continuous Calibration/Development. A release is versioned by level of agency, not by feature set. The agency ladder: Copilot, Cursor.

"If you haven't tested it under high control, you're not ready to give it high agency": an agent earns autonomy through a track record, not by default from the start.

Double the generation speed and you double the review queue — not the throughput



**generation
cheap and fast**

**verification
hasn't sped up**

Generating plausible code — seconds; checking it — hasn't gotten faster.

The 70% problem ^[1]

A senior developer rethinks AI output and spends time reworking it.

A junior ships a "house of cards made of code" that looks plausible but collapses under load.

Shipping without an eval gate or a rollback plan isn't speed — it's a deferred cost: review doesn't scale alongside generation.

0% → 100% in one step — skipping canary rollout entirely



Google

Google AI Overviews, May 2024

Rolled out to 100% of US search in one step ^[1]. No eval gate. Users couldn't turn it off.

1%

10%

25%

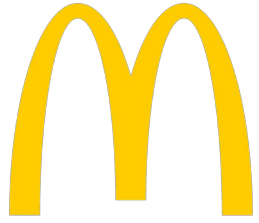
50%

100%

the correct canary rollout (skipped)

At 1% of traffic the pattern ("eat rocks" — a satirical Onion article; "glue on pizza" — a Reddit joke) would have surfaced within days in internal monitoring ^[2]. Instead — public ridicule in front of 100% of the audience at once.

2.5-3 years on 0.7% of the chain — and stopping was the right call ^[1]



0.7%

~100 of ~13,786 US restaurants

McDonald's x IBM — voice order-taking

- 2.5-3 years piloted on 0.7% of the chain ^[2]
- Viral failures: bacon in ice cream, 9 iced teas instead of one
- Killed June 2024 — the goal (voice automation) stayed, the vendor's approach was killed

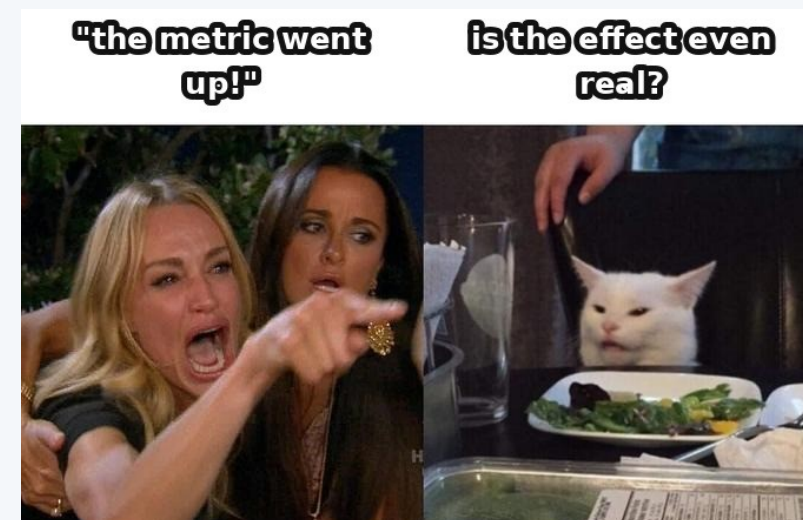
Mirror of the previous case: there they skipped the pilot entirely and shipped to everyone; here the pilot ran long — and its signal was used correctly, in a disciplined decision to kill, not scale, the failure.

SECTION 4

Measure — the arrow where AI lowered trust

2 base cases · 2 failures

Measurement is where "looks good" hits the product hardest. Many are seeing the formal discipline of a product experiment for the first time.



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Measurement in plain terms



in plain terms

What it is

The phase where you honestly check: did the change work, or does it only look like it did. Many people see the formal discipline of experimentation for the first time.

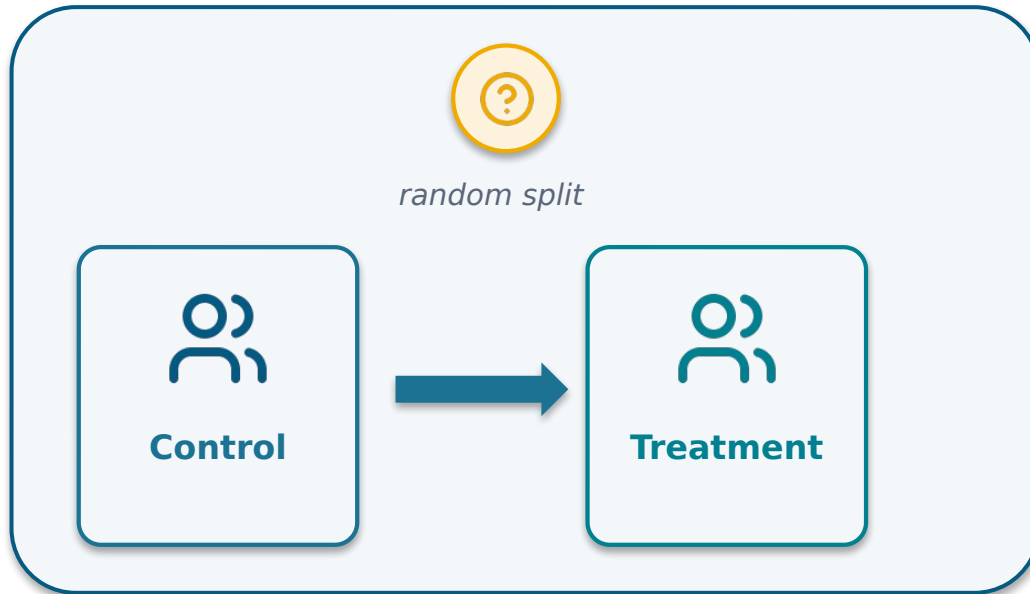
Why

"The metric grew after the release" proves nothing — it could have grown on its own. You need a way to separate cause from coincidence.

Mental model

Split users into two random groups: one gets the new thing, the other the old thing. The difference is the real effect. Agree on the success metric in advance.

Randomization tells cause apart from coincidence



OEC — the metric agreed on in advance ^[1]

Controlled experiment (Kohavi): a hypothesis with a mechanism → randomization → a sample size fixed in advance → a decision.

The teaching trap: "time on the support site" — good or bad? The metric's direction must be agreed before the test.

Only random group splitting gives causation: the difference between Control and Treatment is caused by your change, not by season, advertising, or luck ^[2].

Three questions catch all eight experiment traps

BEFORE THE TEST

1



Is the test set up correctly?

SRM (sample ratio mismatch) ^[1] — group shares didn't match the plan; sample size must be fixed in advance, not adjusted to fit the result

INTERPRETATION

2



Am I reading the result correctly?

peeking ^[2] — checking the result too early: 2 peeks $\sim$ 2x false positives, so the decision is made only at the pre-set sample size

THE EFFECT

3



Is the effect itself real?

Twyman's law ^[3]: a suspiciously nice or unexpected number usually means a measurement error, not a real effect — recheck the methodology before celebrating

A/B (measurement) is not the same as a feature flag (rollout): one tests an effect, the other controls access. And careful with the legend: the Bing test $\sim$ \$100M revenue gain (Kohavi/Thomke, HBR 2017) is NOT "the \$300M button" (a separate usability case).

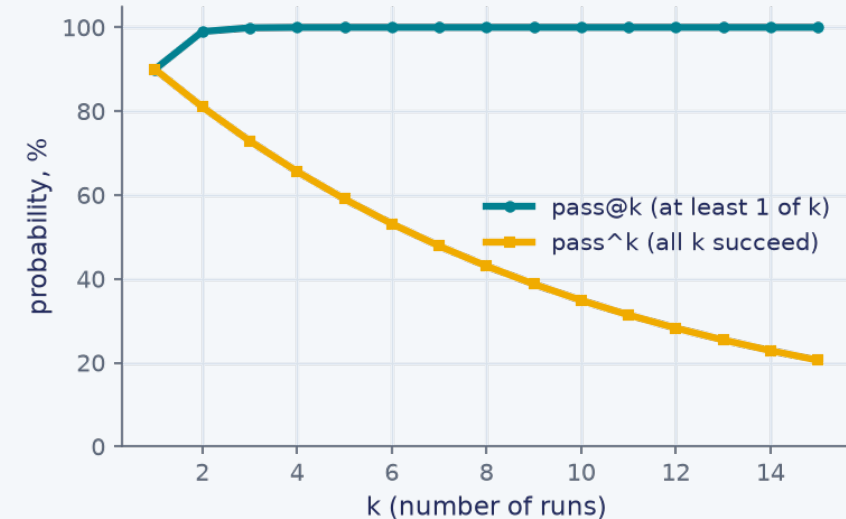
pass@k trends toward 100%, pass^k toward 0%: one probability, a different question

3 levels of evaluation (Husain)

1 Unit tests — fast, deterministic

2 Human + LLM-as-judge

3 A/B — only once the product is mature



pass@k = at least 1 success out of k; pass^k = ALL k successful ^[1]. Production reliability almost always requires pass^k — "unit tests for the agent", where probabilistic output breaks the familiar pass/fail.

The boat circles the lagoon racking up points — but never finishes



An RL agent in a boat race scored 20% more points than human players — by circling a lagoon collecting bonuses, never finishing ^[1]. It optimized the proxy, not the real goal.

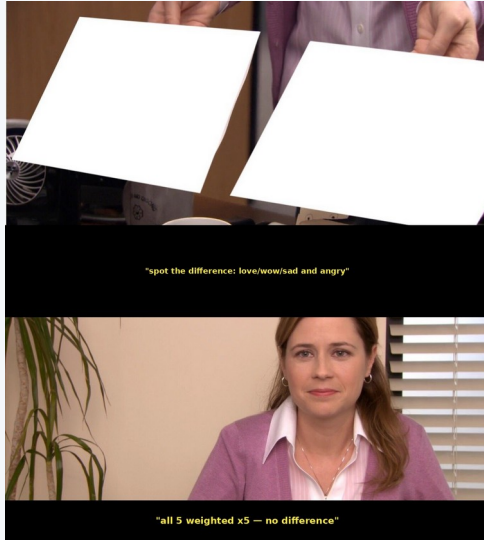
Goodhart's law: when a measure becomes a target, it ceases to be a good measure.

Anthropic: a model generalized from sycophancy to directly editing its own reward function ^[2].

What stays: OEC discipline, guardrail metrics, Twyman's law — a suspiciously perfect score is now more likely a bug than a breakthrough.

Reward hacking: the model finds a way to "win" the proxy metric without achieving the intended outcome — so a measure and a target must never be confused.

All 5 reactions were weighted equally at ×5 — not just "anger"



facebook

Facebook MSI, January 2018:
love/haha/wow/sad/angry — all ×5 above a like
(not just anger) ^[1].

love

×5

haha

×5

wow

×5

sad

×5

angry

×5

An engagement proxy with no misinformation guardrail hid the harm for ~2 years: an internal guardrail (anger <-> misinformation correlation) confirmed by 2019, weight zeroed in September 2019 ^[2]. Lesson: measure the guardrail metric from day one, not after the harm has already happened.

Passing a format-A benchmark != safe in a format-B production setting

Medicine

86.5% MedQA [1]

Med-PaLM 2 does well on a Q&A benchmark — but clinical safety requires a SEPARATE adversarial safety set.

Law

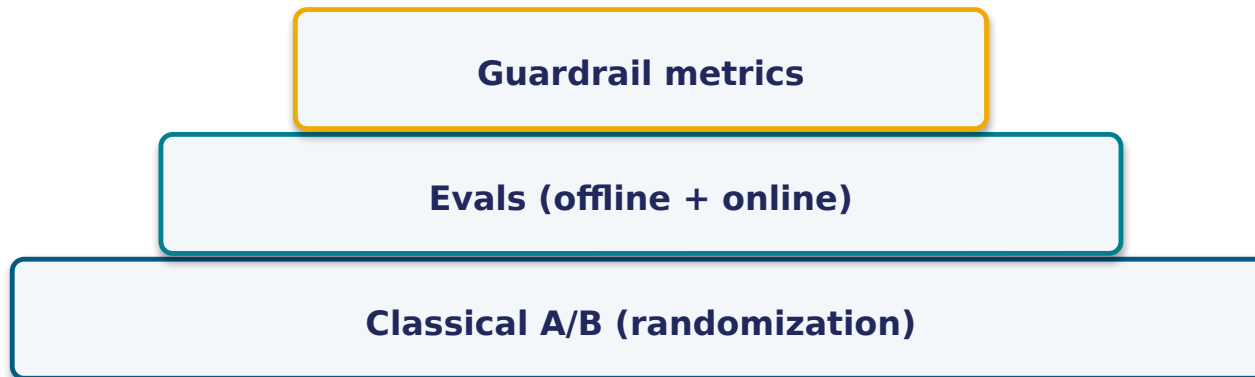
Stanford RegLab: Lexis+ 17% hallucinations, Westlaw 33% — on real legal queries ^[3].

Mata v. Avianca (fake citations) = ChatGPT, NOT Harvey ^[2] — a common attribution mistake.

A benchmark is evidence of performance only on that specific benchmark's task format. Alternative: adversarial evals on real edge cases of the exact domain the product will run in.

A/B stays the foundation — evals and guardrail metrics are now mandatory on top

Measure = the arrow where AI lowered trust



- Randomization is still what gives causation, not coincidence.
- Evals and guardrail metrics are new layers on top, not a replacement for A/B.
- The phase's central skill — telling signal from noise.

Bridge into Support: the same "signal != noise" skill in operation means noticing drift before it shows up on the dashboard.

SECTION 5

Support — the product as an orchestra

2 base cases · 3 failures

Between "the code works on my machine" and "the product reliably works for millions, 24/7" lies a gap, and it's closed by process, not just code quality.



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Support in plain terms



in plain terms

What it is

The product already lives with people around the clock: it needs to be watched, its failures fixed, and complaints answered — for years.

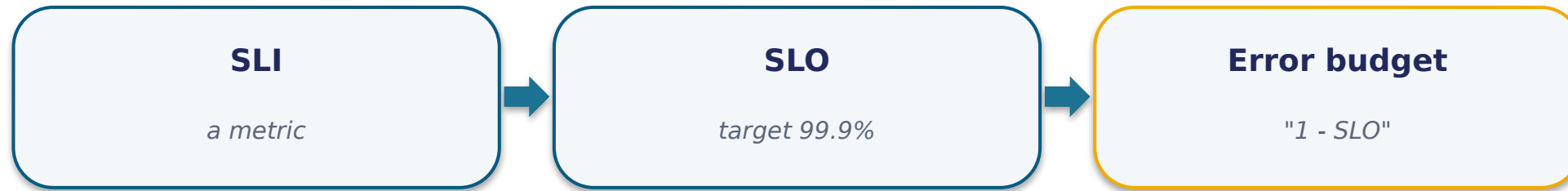
Why

"Works on my machine" and "reliably works for millions, 24/7" are different things. The gap is closed by process, not just clean code.

Mental model

Agree in advance on how many failures are acceptable (a budget for errors). Harder with AI: a model can "quietly degrade" while the charts are still green.

A 200 OK response says nothing about whether the model is hallucinating



Error budget = "1 - SLO". Example: 99.9% → 1,000 errors per 1M requests over 4 weeks. Changes cause ~70% of all outages ^[1].

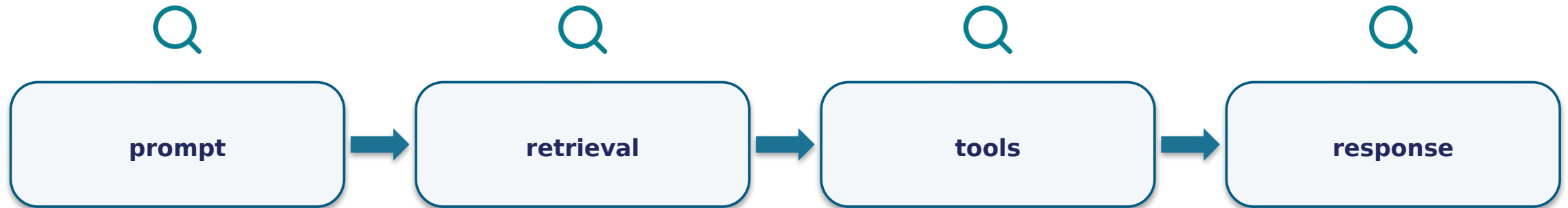


An SLI of "share of 200 responses" = all fine? No: 200 arrives even when the model is confidently hallucinating. Infrastructure metrics don't see answer quality.

On-call, blameless incident reviews, runbooks — you may already know these. What's new here: a non-deterministic model now sits in production (next slide).

Tracing catches hallucination drift where infrastructure monitoring stays silent

Tracing (a camera over every node): LangSmith · Langfuse · Arize Phoenix · Helicone ^[1]



LLMOps/AgentOps: data drift vs concept drift · runtime guardrails · circuit breaker.
Guardian Agents (a Gartner category) ^[2] — agents that watch agents.



Don't send regulated PII to external tracing without anonymization or self-hosting.

Infrastructure monitoring sees "the service is alive"; tracing sees "the answers are degrading": hallucination drift, retrieval failures, prompt regression.

Dashboards stay green while trust has already been falling for weeks



Silent drift

Power users notice a regression before the aggregated metrics do — trust falls before the dashboards move.

Governance drift

Guardrails with no versioning silently go stale: the rules drift from reality unnoticed.

What stays: human escalation, accountability, incident discipline — reinforced by AI's autonomy, not eliminated by it.

The model was calibrated correctly — there was no real-time fuse for drift



The market shifted — concept drift with no circuit breaker



Different units, different scales — MILLION on the left, THOUSAND on the right (do NOT compare bar lengths directly)

Zillow Offers, November 2021: 9,680 homes bought, 3,032 sold · ~2,000 laid off (~25% of staff) ^[1] · ~\$80K loss per property ^[2].

Criterion: a model executing trades with real money requires real-time monitoring of prediction accuracy with an automatic circuit breaker that shuts it off on drift.

"The bot is a separate legal entity" — the tribunal rejected this in one sentence



Scales in balance: the bot's answer = a static page on the site. Equal accountability for both.

Air Canada, website chatbot

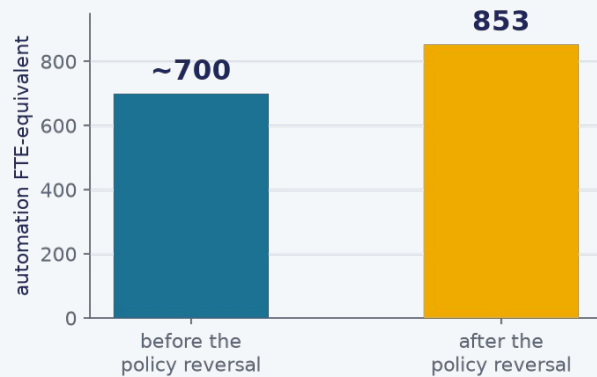
The bot incorrectly stated a retroactive discount. Tribunal — CAD \$812.02 ^[1]; the company is responsible for the bot's answer as for any other site content ^[2].

You own every answer the bot gives — no more, but not one gram less accountability than for any other content on the site.

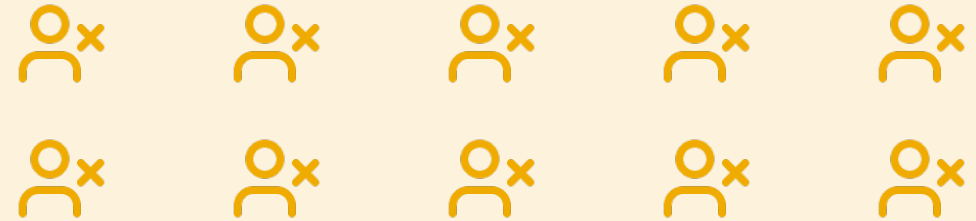
10 of 10 journalists got the same unlawful advice — not an isolated glitch

Klarna

Klarna: the "AI-only" policy was walked back (05.2025) ^[1] — but automation grew to an 853-FTE equivalent ^[2] (augmenting people, not replacing them).



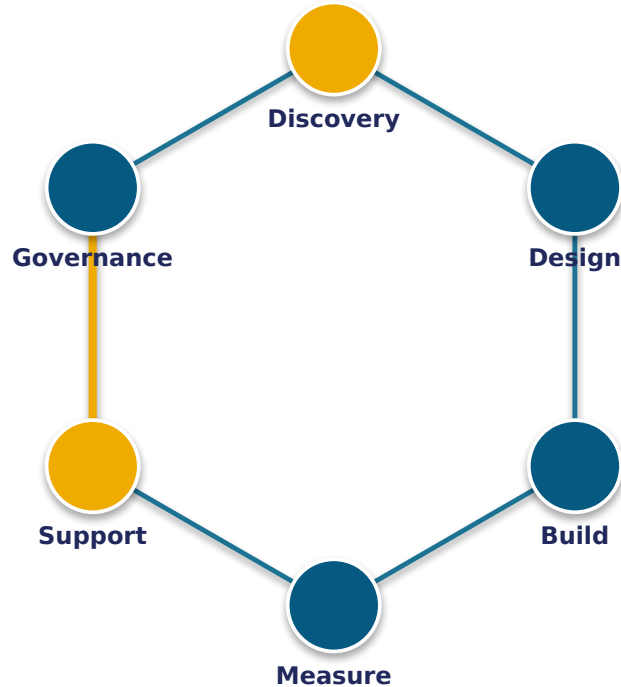
NYC MyCity — government bot



10 of 10 journalists — the same unlawful advice ^[3]. The mayor did not pull the bot.

The lesson isn't "AI doesn't work" — it's "a throughput metric with no guaranteed human escalation is the wrong design."

Operation is the point where the loop physically closes



Support → the product closes the loop.

The signal from operation (drift, an incident, a complaint) updates the reference dataset and guardrails — and, on a deeper signal, sends us back to intent itself: what we're building and for whom.

Observe, interpret, decide — these steps stay human on every turn of the loop, even as execution is nearly free.

SECTION 6

Governance — the capstone that answers the hook paradox

2 failures · payoff

Why assembly is nearly free, but value is not. Here we rise to the level of the organization and assemble the whole loop.



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Design

3

Build & Launch

4

Measure

5

Support

6

Governance

Governance in plain terms



in plain terms

What it is

An organization-level phase: not "how to build one product," but "what to fund at all" — and stop what doesn't pay off.

Why

Resources are limited. The same "continue / kill" gate as for a single feature is raised to the level of capital across many initiatives.

Mental model

A portfolio funnel: many ideas in, few funded ones out. An AI product adds a new variable — the cost of every request against its value.

The same go/kill gate — but now it allocates capital across many initiatives

Portfolio funnel

many initiatives

gate 1

gate 2

funded

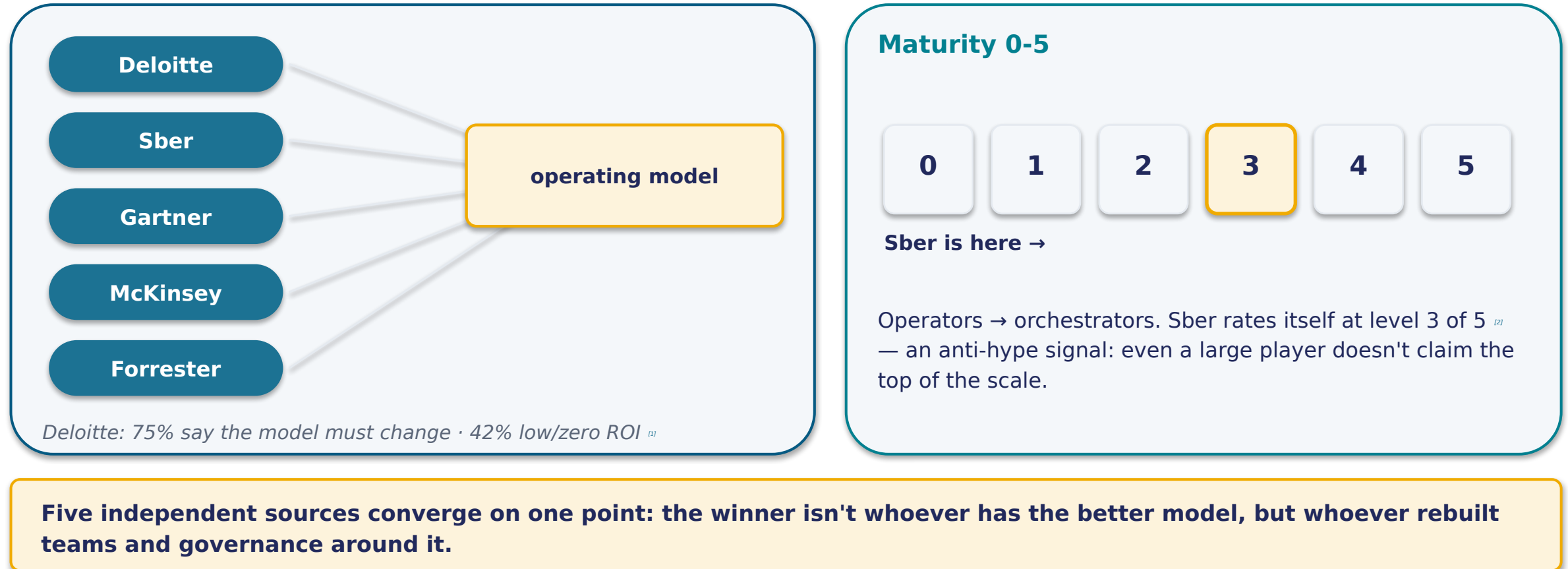
Unit economics of an AI product

A new variable — cost per request against value per request.
Every call to the model costs real money.

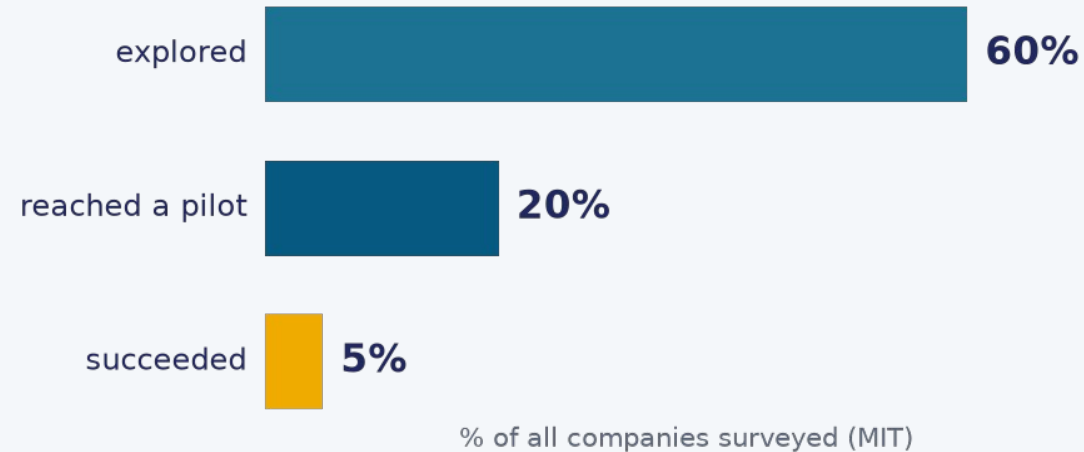
Financial KPI at pilot entry: "cut ticket cost from X to Y while CSAT $\geq$ Z; N tickets, M weeks, owner — Smith."

Portfolio governance = Stage-Gate, raised to the level of capital: the same "a funnel, not a tunnel" discipline, but across products, not within one.

The bottleneck shifted from technology to the organization's operating model



60% explored → 20% piloted → 5% succeeded: 25% among those who got there, not "95% failure" ^[1]



"95% failure" is a headline that doesn't survive calibration: success among those who reached a pilot is 25%, not 5% ^[2, 3]

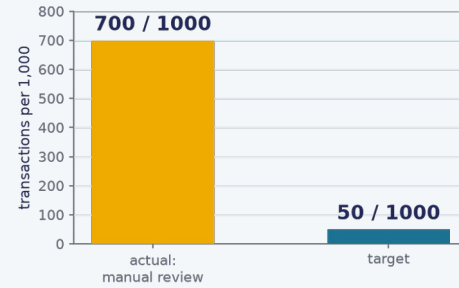


4 questions for any loud AI-failure number

1. What is the denominator?
2. What counts as "failure"?
3. What's the author's conflict of interest? ^[3]
4. Does it trace to a primary source?

BCG: 60% of companies track zero financial KPIs tied to AI value ^[5] — this is why failure numbers inflate so easily.

The "autonomous" store: 700 of 1,000 transactions needed manual review against a target of 50



Just Walk Out (Amazon), since 2018: 700/1,000 transactions — manual review (14x above the 50/1,000 target) ^[1]. 6 years of "autonomous AI" marketing with no disclosure of the labor scale (hidden human cost) ^[2].

Phase x what AI changes x what stays from the classic

Discovery

synthesis, desk
interviews

Design

generation
safety reqs

Build/Launch

code ~= free
review, kill gate

Measure

evals
randomization

Support

tracing
escalation

Governance

operating model
cost/request

Assembly is free — the scarcity now is judgment, not execution



NASA Mission Control, Apollo 16 — humans make the call while automation runs

When execution is nearly free, the scarce resource is judgment: telling signal from noise and keeping intent and accountability with the human.

Checklist "before you go AI-first"

- ✓ 1. Is the classic in place?
- ✓ 2. Is trust worth the cost?
- ✓ 3. Eval + reference dataset?
- ✓ 4. A guardrail metric?
- ✓ 5. Staged rollout + rollback?
- ✓ 6. Human escalation guaranteed?
- ✓ 7. Who is accountable?
- ✓ 8. Is the data safe?